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Documentation: Water Bodies Segmentation Project

Introduction

This project focuses on water bodies segmentation, employing three distinct models to achieve accurate segmentation results. Two models were developed from scratch, while one pre-trained model was incorporated to compare performance and efficiency. The primary objective was to identify and segment water bodies in images, providing valuable insights for environmental monitoring and resource management.

Objectives

1. Develop two custom models for water bodies segmentation.
2. Utilize a pre-trained model to benchmark against the custom models.
3. Compare the performance of the models based on accuracy, precision, recall, and other relevant metrics.

Methodology

Dataset

- Image Directory: E:/Msa/pattern recognition/Project/Water Bodies Code/Water Bodies Dataset/Images
- Label Directory: E:/Msa/pattern recognition/Project/Water Bodies Code/Water Bodies Dataset/Masks

The dataset consists of images and corresponding masks representing water bodies.

Models

1. U-Net

- General Details: U-Net is a widely used architecture for image segmentation tasks, known for its encoder-decoder structure and skip connections that help preserve spatial information.
- Architecture Details: The encoder downsamples the input using convolutional and pooling layers, while the decoder upsamples to produce segmentation

masks. Activation functions include ReLU for hidden layers and sigmoid for the output layer.

- Training: The model was trained using the Adam optimizer with a learning rate of 3e-4 for 100 epochs. Early stopping and checkpoint callbacks were used to prevent overfitting.

2. SegNet

- General Details: SegNet is a deep convolutional neural network designed specifically for pixel-wise segmentation. It features an encoder-decoder architecture that uses max-pooling indices for more efficient upsampling.
- Architecture Details:
    - Encoder:
        - Input size: (128, 128, 3)
        - Convolutional layers with ReLU activation
        - Max-pooling with indices
    - Decoder:
        - Upsampling layers using max-pooling indices
        - Convolutional layers with ReLU activation
        - Final layer: Conv2D with sigmoid activation
- Training: The Adam optimizer with a learning rate of 3e-4 was used. ReduceLROnPlateau and other callbacks ensured efficient training. Example code:
- reduce_lr = ReduceLROnPlateau(
-   monitor='val_loss',
-   factor=0.5,
-   patience=5,
-   min_lr=1e-7
-)

3. EfficientNet

- **General Details: EfficientNetB0, pre-trained on ImageNet, was adapted for segmentation tasks. Its efficient block design enables high performance with fewer parameters compared to traditional models.**
- **Architecture Details:**
  - **A pre-trained EfficientNetB0 encoder integrated with a U-Net decoder.**
  - **Input shape: (128, 128, 12) channels, mapped to 3 channels using a 1x1 convolution.**
  - **Depthwise separable convolutions and swish activations for efficiency.**
  - **Decoder layers ensure high-resolution segmentation outputs.**
- **Training: Fine-tuned with the Adam optimizer, using a learning rate of 1e-4 for 100 epochs. ReduceLROnPlateau was used to adjust learning rates dynamically. Example training code:**
- **history = model.fit(train_images,**
- **train_labels,**
- **validation_data=(val_images, val_labels),**
- **epochs=100,**
- **batch_size=32,**
- **callbacks=[early_stopping, checkpoint, reduce_lr])**

Results

Loss and Validation Metrics

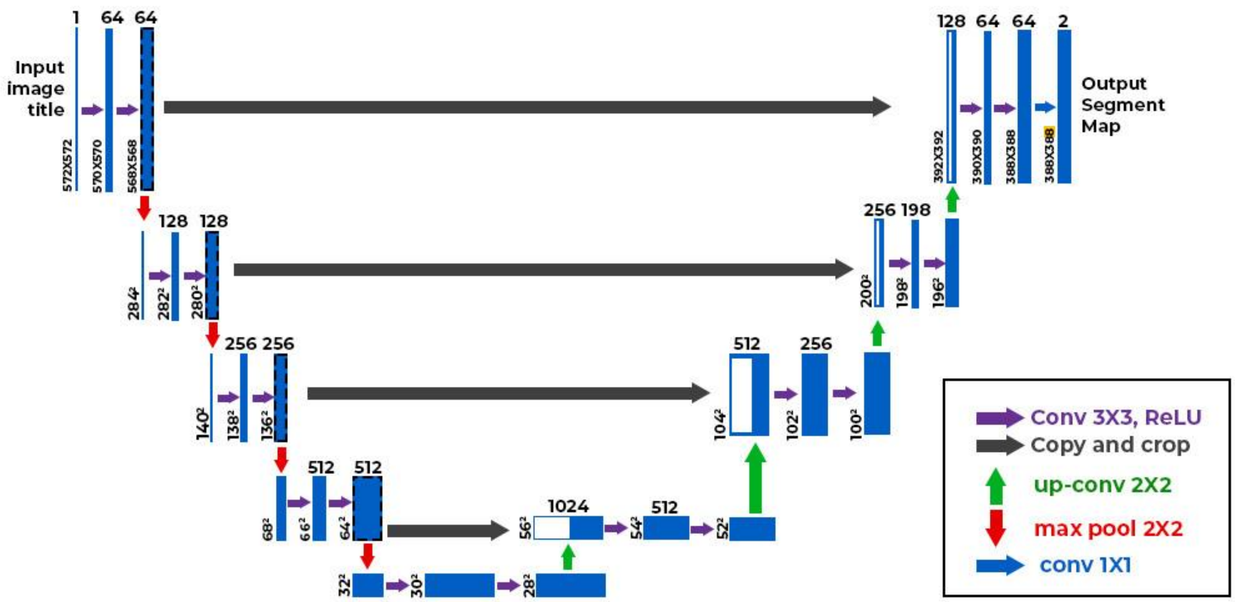
Model	Loss	Val Loss	Val Accuracy	Learning Rate
U-Net	0.3433	0.3440	0.9241	3.0000e-04
SegNet	0.2899	0.2983	0.7493	3.0000e-04
EfficientNetB0	0.1952	0.5100	0.7363	1.0000e-04

Visual Comparisons

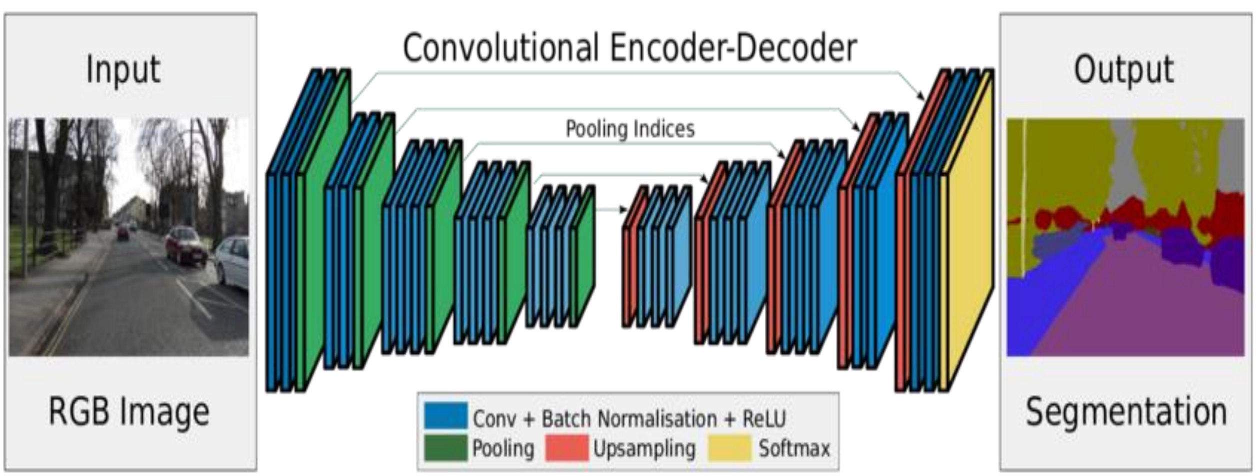
- **Add visual comparisons of predictions from each model alongside ground truth masks.**

Figures

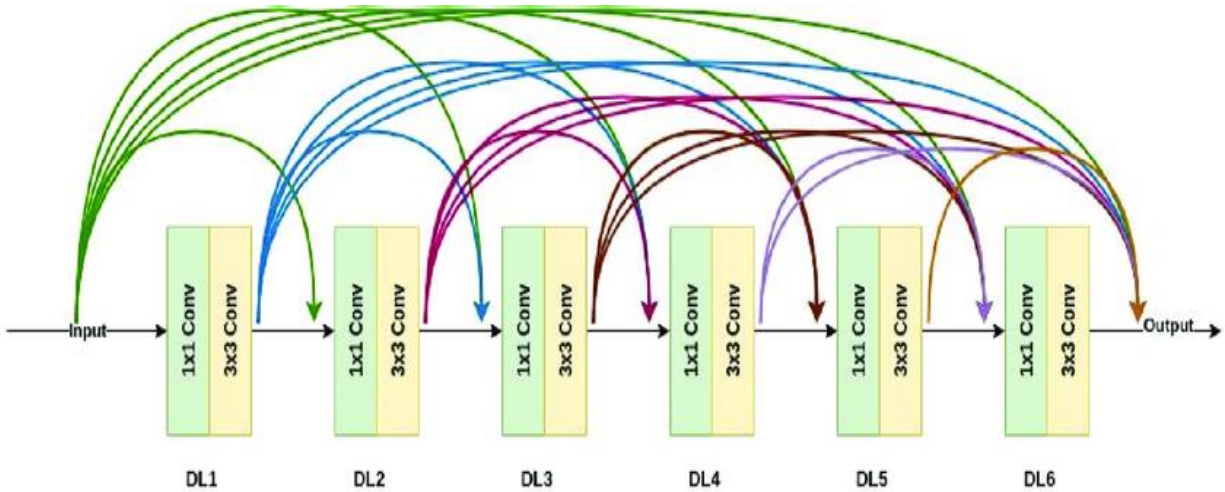
U-Net Architecture



SegNet Architecture



EfficientNet Architecture



Conclusion

This project demonstrates the potential of U-Net, SegNet, and EfficientNet models in achieving accurate water bodies segmentation. The integration of diverse approaches provides a comprehensive evaluation of segmentation strategies for environmental applications.

References

https://doi.org/10.1016/j.asoc.2022.109038

https://doi.org/10.1038/s41598-024-67113-7